NIST-Aligned Credit XAI & Bias Mitigation Engine

Bridging the Gap between AI Policy and Technical Implementation

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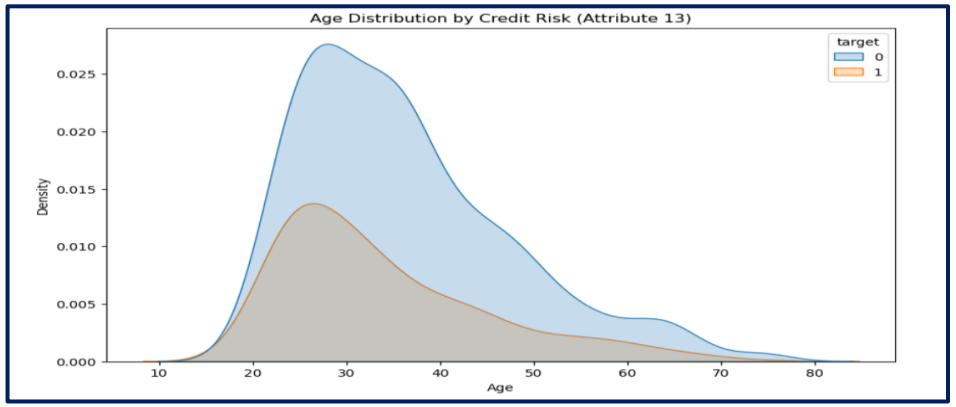
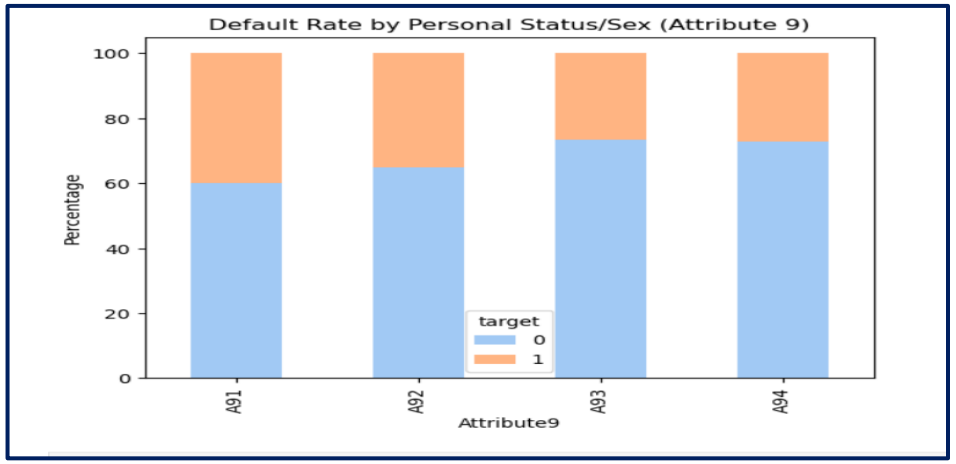
- **Focus:** A “Living Lab” Case Study on the NIST AI RMF & ISO/IEC 42001

The Compliance Paradox: Hidden Bias in Protected Groups

Why Removing "Age" isn't Enough to Ensure Fairness

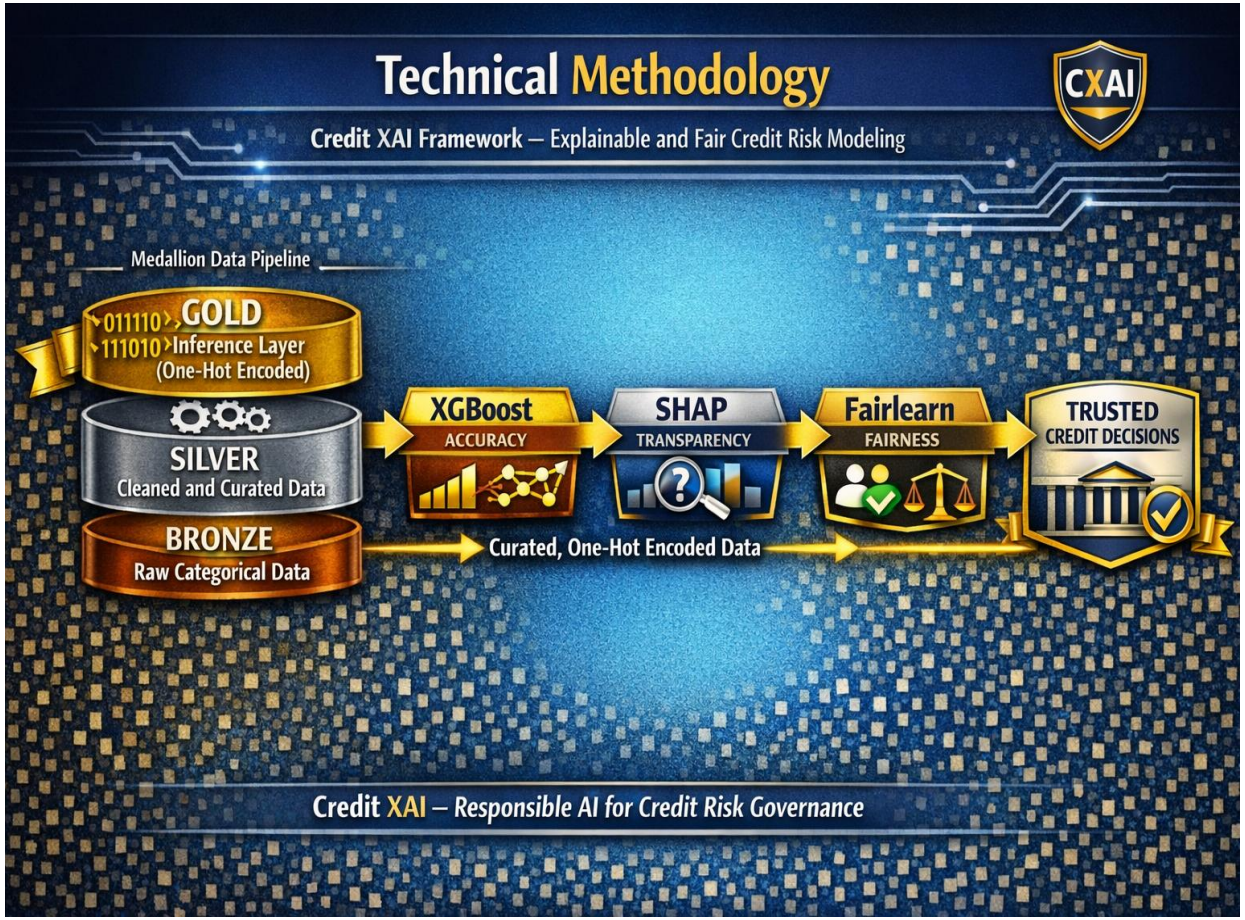
- **Problem:** Initial audit revealed a 0.73 Group Parity Ratio against younger applicants (<25), despite removing direct protected attributes.
- **Root Cause:** Identifying "Proxies" like **Attribute 9** (Personal Status) and **Attribute 15** (Housing) that leak demographic info into the model.
- **Goal:** Move from "Blindness" to "Technical Fairness"





Technical Architecture & The Medallion Data Pipeline

Building a Robust “Bronze to Gold” inference Environment



- **Stack:** Python, XGBoost, Scikit-learn, and Streamlit.
- **Data Integrity:** Transitioning raw categorical data into 47-feature One-Hot Encoded “Gold” Layer for model consistency.
- **Audit Tooling:** Utilizing **Fairlearn** for disparity metrics and **SHAP** for local/global explainability.

The “Hero” Metrics: Accuracy vs. Fairness

Achieving Regulatory Compliance Without Sacrificing Performance

- **Fairness Improvement:** Boosted Group Parity from **0.73** to **0.9583** using a ThresholdOptimizer strategy.
- **Performance Stability:** Maintained a 0.7992 ROC-AUC (negligible variance of -0.0008).
- **Impact:** Proved that a “Fair” model can remain a “High-Performing” model.

```
--- FINAL GOVERNANCE VALIDATION ---
                Model ROC-AUC Fairness Ratio Status
Compliant (Biased)  0.7992         0.7324 ❌ FAIL
  Optimized (Fair)  0.7992         0.9583 ✅ PASS
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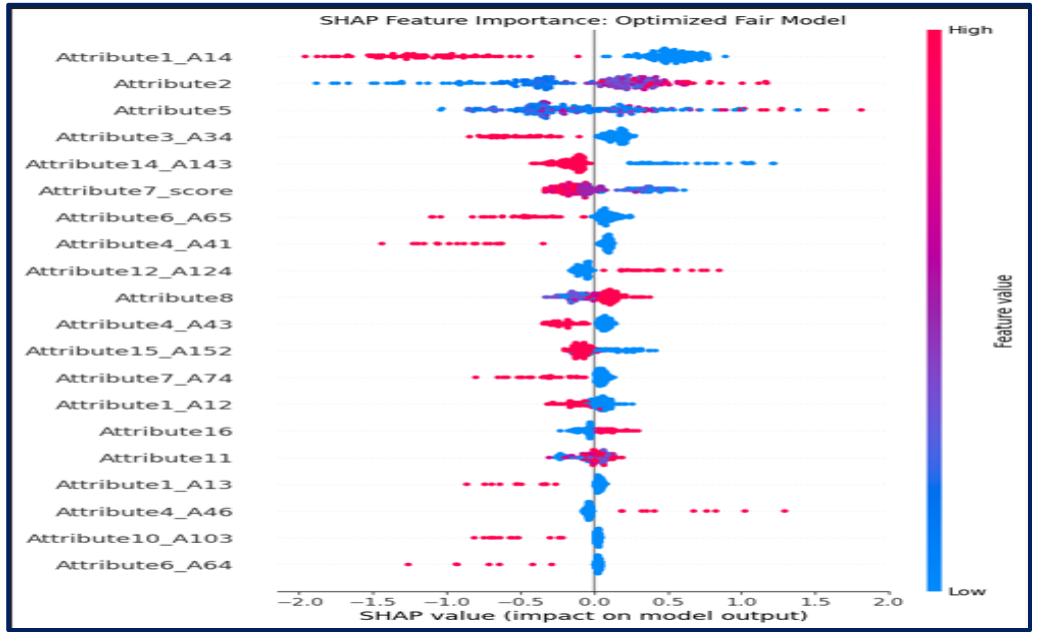
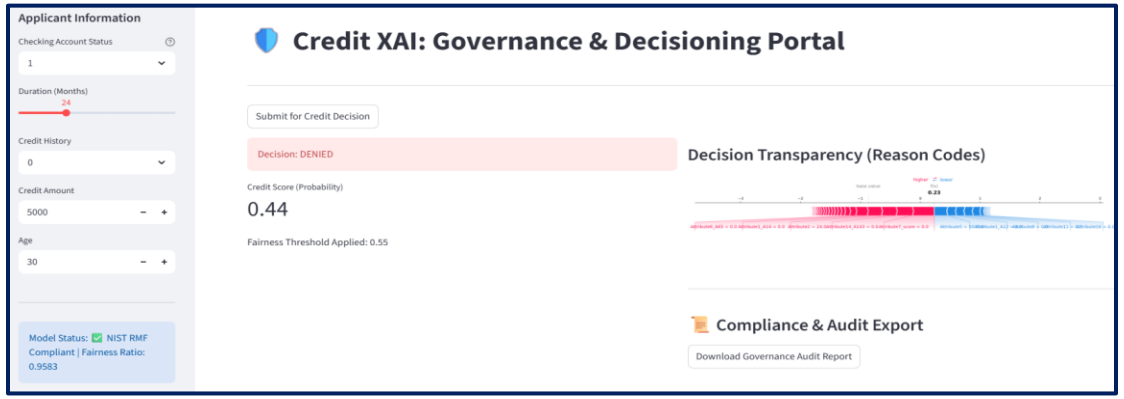


Opening the Black Box: Reason Codes & Audit Trails

Empowering Loan Officers with Explainable AI (XAI)

- **Local Explanation:** Real-time SHAP Force Plots providing “Reason Codes” for every decision.
- **Human-in-the-loop:** A Streamlit dashboard that allows loan officers to understand the why (e.g., Checking Account Status vs Duration).
- **Governance Output:** Automated generation of timestamped Compliance Audit Reports (.txt) for internal records.

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CREDIT XAI GOVERNANCE AUDIT REPORT
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Timestamp: 2026-04-17 15:05:27.059048
Model Version: Certified Compliant XGBoost (v1.0)
NIST Pillar: Manage (Phase IV/V)

APPLICANT DATA SUMMARY:
- Age: 30
- Requested Amount: 5000
- Duration: 24 months

GOVERNANCE DECISION:
- Probability Score: 0.3737
- Applied Fairness Threshold: 0.55
- Final Decision: DENIED

FAIRNESS VALIDATION:
- Protected Attribute: Age
- Mitigation Strategy: ThresholdOptimizer (Post-processing)
- Group Parity Ratio: 0.9583 (PASSED)

EXPLAINABILITY (SHAP):
- Primary Decision Driver: Attribute 1 (Checking Account Status)
- Governance Note: Model successfully decoupled creditworthiness from age-proxy features.
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END OF REPORT
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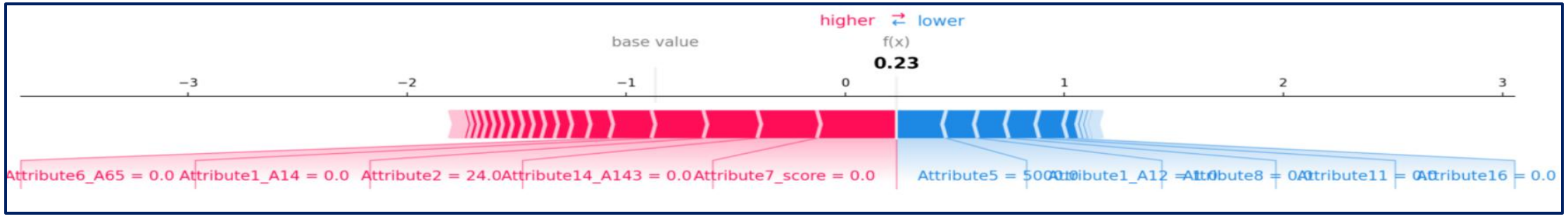

Scaling Responsible AI Governance

From Theoretical Frameworks to Functional Products

- **AIGP Body of Knowledge Alignment:** Directly applied IAPP AIGP standards for algorithmic accountability and transparency to bridge the gap between AI policy and technical deployment.
- **Key Takeaway:** AI Governance is a technical discipline, not just a policy requirement.
- **Future Scope:** Expanding the “Living Lab” to include drift monitoring and adversarial robustness.
- **Call to Action:** “View the full Technical Audit & Repository on GitHub” https://github.com/vrrgithub1/credit-xai-governance/